AFRINEXT · PHASE 1 · FOR SENIOR DEVELOPER REVIEW

Phase 1 Review Packet

Everything produced so far, how to verify it independently, what was and was not tested, and the decisions that need a sign-off before Phase 0 starts.

Branch **claude/create-app-repository-1cxsih** | Head **13cc87d** | 2 commits | Build **green** | Lint **clean** | Automated tests **none**

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1 What exists

Two deliverables, with **very different review weight.** Read this before allocating time — most of the code in the repository is deliberately temporary, and reviewing it carefully would be wasted effort.

BLUEPRINT — DECISIONS THAT ARE EXPENSIVE TO REVERSE	CODE

Suggested split of a ~2 hour review: ~1h45 blueprint · ~15 min code skim

#	DELIVERABLE	WHERE	SIZE	STATUS
D1	**Phase 1 technical blueprint** Stack, schema, ledger design, permissions, security, roadmap	commit 13cc87d docs/architecture/	~9 000 words 3 diagrams	NEEDS DEEP REVIEW Architecture only — no code written from it.
D2	**Working prototype** Mobile-first PWA directory: browse, search, detail, submit, saved, offline	commit 0af2da2 src/	2 243 LOC 9 routes	SKIM ONLY Runs and is verified, but ~70% is replaced in Phase 0.

WHY D2 IS NOT THE FOUNDATION

D2 was built to answer "can we create an app in this repository?" before the SaaS brief arrived. It has **no database, no authentication, no payments and no money** — listings live in an in-memory object that resets when the server restarts. What survives into Phase 0 is the UI layer (design tokens, PWA shell, mobile components) and two patterns worth keeping: URL-driven search state, and server-side revalidation of every submission. The data layer is discarded.

The blueprint is published as a browsable document here: Afrinext Technical Blueprint. The same file is in the repository at `docs/architecture/blueprint.html`, with a Markdown decision index beside it at `docs/architecture/README.md` — that index is the fastest way in, and is the thing worth diffing as decisions change.

2 Verify it yourself

Nothing below needs credentials, an account, or network access beyond npm and GitHub. Roughly ten minutes end to end.

1 Check out the branch

```
git clone https://github.com/abdoulasani/Afrinext.git
cd Afrinext
git checkout claude/create-app-repository-1cxsih
git log --oneline main..HEAD   # expect exactly 2 commits
```

2 Install

Node 22 or newer. No postinstall scripts, no native builds.

```
npm install
```

3 Typecheck, lint and build

`next build` runs the TypeScript check, so a green build is also a green typecheck. Both should exit zero with no warnings.

```
npm run lint
npm run build
```

Expected: `Compiled successfully`, `Finished TypeScript`, and a route table listing 9 routes.

4 Run it and exercise the API

```
npm run start &                       # serves the production build on :3000

# search, filter and sort
curl -s 'localhost:3000/api/listings?q=shea'
curl -s 'localhost:3000/api/listings?category=logistics&verified=1'
curl -s 'localhost:3000/api/listings?sort=newest'

# server-side validation — expect HTTP 422 with per-field errors
curl -i -X POST localhost:3000/api/listings \
  -H 'Content-Type: application/json' \
  -d '{"name":"X","tagline":"short","category":"nope","contact":{}}'

# malformed body — expect HTTP 400
curl -i -X POST localhost:3000/api/listings \
  -H 'Content-Type: application/json' -d 'not json'

# unknown listing — expect HTTP 404
curl -i localhost:3000/api/listings/does-not-exist
```

5 Open it on a phone-sized viewport

`localhost:3000` at 390×844. Check the bottom tab bar, the browse filters, a listing page, and the submit form. Bookmark a listing and confirm it appears under *Saved* — that state is `localStorage`, per device, by design.

6 Read the blueprint

Start with `docs/architecture/README.md` — thirteen decisions in a table, then the open questions. Follow into the full document only where a decision looks wrong.

3 What was tested, and what was not

Stated plainly so the review starts from an accurate baseline rather than an assumption.

Verified, against a running production build

CHECK	RESULT
Production build and TypeScript check	PASS 9 routes emitted
ESLint	PASS clean, zero warnings
Search, category/country filter, verified filter, three sort orders	PASS correct result sets
Invalid submission	PASS HTTP 422 with per-field errors
Valid submission	PASS HTTP 201, appears in search, renders its own page
Duplicate business name	PASS unique slug generated (`…-2`)
Malformed JSON body	PASS HTTP 400
Unknown listing id	PASS HTTP 404 on both API and page
Render at 390×844 (Chromium)	PASS no console errors, no horizontal overflow on any screen
Save → Saved round trip	PASS persists across navigation

NOT TESTED — ASSUME NOTHING HERE WORKS UNTIL PROVEN

- **There is no automated test suite.** Every result above came from driving the running app by hand. Nothing guards against regression. Vitest and Playwright are scheduled in Phase 0.
- **No browser other than Chromium**, and no real device — desktop Chromium at a phone viewport is not the same as a mid-range Android on a 3G connection.
- **No load, soak or concurrency testing.** The in-memory store is not concurrency-safe and was never intended to be.
- **No security review.** The prototype has no authentication, so every API route is open by design — including `POST /api/listings`, which anyone can call. Acceptable for a demo, disqualifying for anything public.
- **No accessibility audit** beyond semantic markup, focus states and labelled controls.
- **Nothing from the blueprint is implemented.** No database, no ledger, no auth, no payments. The document describes work not yet begun.

4 Review checklist — the blueprint

These are the questions worth an opinion. Each names the decision, why it was made, and the specific thing that would make it wrong. A "no" on any row is more valuable than a "looks fine" on all of them.

Money — the decisions that cannot be retrofitted

DECISION	QUESTION FOR YOU
Double-entry ledger, immutable entries, per-transaction balancing enforced by a deferred database trigger	Is the invariant set complete? Specifically: is a deferred `CONSTRAINT TRIGGER` the right enforcement point, or would you prefer the balance check inside a stored procedure that owns all posting?
Pending and available are two separate ledger accounts; settlement is a transfer between them	Agree, or would you model it as a status on the transaction? The claim is that two accounts make "why is my money still pending" answerable by pointing at a row.
Balances are a cache in `account_balances`, verified nightly against the sum of entries	Is nightly frequent enough? Should drift detection block payouts automatically, or only alarm?
Money is `bigint` minor units + currency code; the exponent comes from a `currencies` table because XOF and XAF have zero decimals	Any place in the plan where this leaks — API responses, CSV exports, the admin console?
Rounding: non-seller lines round half-up, the seller gets `gross − Σ(other lines)`	Does the residual-to-seller rule hold up commercially, and is it defensible if a seller queries it?
Commission rules resolved once at order creation and frozen in `order_fee_lines`	Is the resolution order (most-specific-wins, then priority, then `effective_from`) unambiguous enough to avoid two rules tying?

Stack — where I want a second opinion

DECISION	QUESTION FOR YOU
CONTESTED Drizzle ORM over Prisma	**Your call, and it depends on the team.** Drizzle for SQL-first migrations and native CHECK constraints; Prisma for migration tooling and DX on a 40-table schema. What does the team actually know?
pg-boss over BullMQ + Redis	The argument is transactional enqueue — a settlement job commits with the ledger rows it will act on. Do you accept the throughput ceiling that comes with it?
Domain logic in a framework-free `packages/core`, enforced by an ESLint boundary rule	Is the boundary drawn in the right place? Anything that will inevitably want to import from Next.js?
Database sessions rather than JWTs; phone OTP as the primary credential	Agree on revocability over statelessness? Any concern about session-table load at scale?
Postgres in `eu-west-3` (Paris) rather than `af-south-1`	UNVERIFIED This is an assumption about routing, not a measurement. Do you have real latency numbers from the launch cities?

Model and scope

DECISION	QUESTION FOR YOU
Narrow the first launch to digital products and courses, one country, one payment rail	~18 weeks instead of ~32. Does dropping physical goods and delivery from launch cost more commercially than the schedule saves?
Build the ledger in P1, before any product feature	Three weeks with nothing visible to show. Is that defensible to the business, and would you sequence it differently?
Referral is single-level, paid only from settled transactions, basis is platform revenue not gross	The bound is structural — the platform can never owe more than it earned. Does single-level cost too much growth to accept?
Roles are scoped rows; no `role` column on `users`; IDOR prevented by scoping queries rather than post-fetch checks	Does the `authorize(actor, permission, scope)` choke point cover every case you can think of?
Paid video protected by short-TTL signed tokens and a viewer overlay, with DRM explicitly out of scope	Is that honest tier acceptable to tell instructors, or does the business need real DRM at launch?
iPay behind a provider interface, with `createPayout` optional; a manual transfer rail closes the economic cycle in P3	Anything in the interface that will not survive contact with the real API?

OUTSIDE ENGINEERING, BUT IT GATES THE SCHEDULE

The blueprint flags that holding and disbursing user funds may be a licensed activity under BCEAO rules in the UEMOA zone, with separate regimes in Nigeria and Ghana. That is a legal question, not an architectural one — but it gates Phase 6 (payouts) and could reshape the wallet's legal model. If you know the answer or know who does, that unblocks the largest open risk in the plan.

5 Review checklist — the code

Fifteen minutes. The question is not "is this production quality" — it is not, and it is not meant to be. The question is **whether the patterns worth keeping are the right ones.**

FILE	LINES	WHAT TO LOOK FOR
`src/lib/store.ts`	64	The storage boundary. Everything that reads or writes goes through here, so the module can be swapped for Drizzle without touching callers. **Is the surface right?** If it is wrong, it is wrong in Phase 0 too.
`src/lib/search.ts`	109	Search state lives entirely in URL params, so results are linkable and the back button works. Pattern carries forward; the implementation moves to SQL.
`src/lib/validate.ts`	104	Server always revalidates; client checks are a convenience. Becomes Zod schemas shared across HTTP, server actions and jobs.
`src/app/api/listings/route.ts`	32	Status-code discipline: 201 / 400 / 422 / 404. Note there is **no authentication** — deliberate for a demo, and exactly what Phase 0 fixes.
`src/lib/useSaved.ts`	74	`useSyncExternalStore` over `localStorage`, with a hydration guard so saved state never flashes on SSR. Replaced by server-side favourites once accounts exist.
`src/components/`	~700	Design tokens, mobile shell, safe-area handling. This is the part that survives into `packages/ui`.

Worth **not** spending time on: `src/lib/seed.ts` (310 lines of fixture data, deleted in Phase 0), the visual design (subject to change once there is a real product), and anything about how listings are ranked (the ranking function is a placeholder).

6 Sign-off sheet

Four items block Phase 0. Six can be settled during it. Nothing starts until the four are answered — an approval that skips them would be approving an unknown.

Blocking

#	QUESTION	MY RECOMMENDATION	WHO DECIDES
B1	Who is advising on whether Afrinext may hold and disburse user funds in the launch country?	Engage local counsel before Phase 6. Structure so funds sit with the licensed PSP and Afrinext records entitlement.	Business / legal
B2	Launch scope — digital and courses first, or physical goods at launch?	Digital and courses. ~18 weeks vs ~32, and it de-risks the hardest parts first.	Business
B3	Is the iPay contract signed, and can we get the documentation and sandbox credentials?	If not, the manual transfer rail from P3 becomes the launch payment method — plan for it openly rather than waiting.	Business
B4	Referral — single level only?	Yes. Tiers materially increase regulatory exposure and contradict the brief's own principle.	Business, with legal input from B1

Settle during Phase 0

#	QUESTION	DECIDES
S1	Team size and Prisma vs SQL experience	The ORM choice — the only reason it is still open
S2	Launch country and currency (Niger / XOF assumed)	Seed data, locale default, payment channels
S3	Hosting — managed (Vercel + Neon) or containers (Hetzner / Fly)	Deployment shape and unit cost
S4	Is the PWA sufficient, or is a store-distributed app required?	How much weight the REST layer carries early
S5	Cash on delivery at physical-goods launch?	Whether the rider cash-float model is in scope
S6	KYC — which documents are verifiable locally, and is there an existing provider?	Withdrawal tiers in Phase 6

RECORD THE VERDICT

Per item: APPROVED | APPROVED WITH CHANGES | REJECTED — and for anything other than a clean approval, the reason. I will revise the blueprint against the feedback and reissue it before any code is written, rather than starting and patching later.

7 What happens after approval

The gate is real, not ceremonial. Phase 0 writes the database schema and the permission model — both are load-bearing for everything after, and both change shape depending on the four blocking answers in section 6.

Phase 0 — foundations · 2 weeks

Turn the repository into the monorepo the blueprint describes, and stand up the parts every later phase depends on.

- pnpm workspaces + Turborepo; move today's app into `apps/web`, extract components into `packages/ui`.
- Postgres, Drizzle schema v1 (or Prisma, pending S1), forward and rollback migrations in CI.
- Authentication: phone OTP and email, database sessions.
- Scoped RBAC with the single `authorize()` choke point.
- i18n with French default; `packages/i18n` shared with the worker.
- CI: typecheck, lint, test, migrate up and down on every commit. Vitest and Playwright wired up — **this is where the missing test suite gets fixed**.
- Observability skeleton: structured logs with request and actor ids.

Exit criterion: a user registers by phone, signs in, switches language; CI is green on typecheck, lint, tests and migrations in both directions.

Phase 1 — the money spine · 3 weeks

Nothing visible ships. This is the part that cannot be retrofitted, so it is built before anything depends on it.

- `packages/core/money` — Money type, currency minor units, the rounding rule.
- `packages/core/ledger` — posting, immutability, the balance trigger, settlement holds.
- Commission engine with rule resolution and frozen fee snapshots.
- Idempotency keys on every money-moving operation.

- Nightly reconciliation job with drift alarms, and an admin ledger viewer.

Exit criterion: property-based tests over 10 000 randomised operations prove every transaction balances and the cached balance equals the sum of entries; an injected drift is caught by the reconciliation job. Your senior developer should be able to read that test file and agree it proves what it claims.

> **WHAT THE NEXT PACKET WILL CONTAIN**
>
> Phase 0 hands over a running monorepo, a migration you can run against your own Postgres, and a CI run to inspect. Phase 1 hands over the ledger with its property tests — and that one is the most important review in the whole project, because a ledger bug is discovered by users, in money.

8 Standing handoff format

Every phase from here produces a packet in this shape, so the review is the same shape each time and nothing depends on remembering what happened last month.

SECTION	CONTENTS
1 · What exists	Deliverables, commit hashes, size, and where to spend review time
2 · Verify it yourself	Copy-pasteable commands, no credentials needed, expected output stated
3 · Tested / not tested	Every check run and its result — and an explicit list of what was *not* covered
4 · Review checklist	Targeted questions, not "please review". Each names what would make the decision wrong
5 · Exit criterion	The phase's stated criterion, and the evidence that it passed
6 · Sign-off sheet	Blocking items separated from ones that can be settled in flight
7 · What happens next	Next phase scope and its exit criterion, so approval is informed

Two commitments that make the packet worth trusting: **nothing is reported as working unless it was run**, and anything unimplemented, unverified or assumed is labelled as such rather than left to inference. Where a check was skipped, the packet says so — see section 3 of this one, which lists six things that were not tested.

Prepared for review · branch `claude/create-app-repository-1cxsih` · head `13cc87d`

AFRINEXT · PHASE 1 DELIVERABLE

Afrinext Technical Blueprint

The architecture for a multi-country African commerce and learning platform built around a double-entry ledger — stack, schema, permission model, money design, security controls and a phased build order.

Status: for review | Scope: architecture only | Code written: none | iPay: not integrated

CONTENTS

00 The read, and one pushback

Afrinext is three businesses sharing one identity system. A **marketplace** (physical goods, digital goods, services), a **learning platform** (paid video and documents with entitlement), and a **money system** (wallet, commissions, settlement, payouts). Delivery and the referral network are not separate products — they are two more sources of ledger events.

That framing decides the architecture. The ledger is the spine; everything else is a producer of financial events that the ledger records. A course purchase, a delivered parcel and an ambassador commission are the same shape of thing once they reach the money layer. If that layer is designed last, every module above it grows its own private notion of "balance" and the platform becomes unauditable.

The "one person, one account, many roles" principle is right and it is cheap to honour, provided one rule holds from the first migration: **there is no `role` column on the user.** Roles are rows, they are scoped to a resource, and they are additive.

The pushback

The module list in the brief is a two-to-three-year product. Built as specified — all modules in parallel — every part arrives shallow, and the hardest, least reversible part (money) gets exercised last, by which point the schema is load-bearing for a dozen features.

RECOMMENDATION

Launch one vertical: digital products and courses, one country, one payment rail.

WHY	It exercises identity, roles, stores, catalog, checkout, payments, ledger, commissions, wallet, withdrawals, entitlement and protected video — every hard part except logistics — with no inventory, no couriers, no cash handling and near-instant fulfilment. It is the shortest path to a real settled transaction, which is the only thing that proves the platform works.
INSTEAD OF	Launching the full marketplace, delivery network and creator marketplace together.
TRADE-OFF	Physical goods and delivery slip to a second launch, roughly five weeks of work that starts later rather than never. Ambassadors have a thinner catalog to promote at day one.
REVERSIBLE?	Yes. Nothing in this blueprint's schema or module boundaries assumes the narrow scope — the tables for physical goods and delivery are designed here, just not built first.

This is a recommendation, not an assumption I have baked in. If you want physical goods at launch, say so and I will reorder the roadmap in section M — the architecture does not change, only the build order.

What is in the repository today

The current `Afrinext` repository holds a mobile-first PWA directory prototype: Next.js 16, Tailwind v4, an in-memory listing store, seed data, and a submission form. It is honest about what it is — there is no database, no authentication, no payments, and no money. In this plan:

ASSET	DISPOSITION
Design tokens, PWA shell, mobile navigation, card and form components	CARRIES FORWARD Becomes the seed of `packages/ui`.
Next.js 16 + TypeScript + Tailwind v4 setup, lint and build config	CARRIES FORWARD Becomes `apps/web` inside the monorepo.
`src/lib/store.ts`, `src/lib/seed.ts`, the in-memory model	DISCARD Replaced by Postgres and Drizzle.
`src/lib/search.ts`, `validate.ts`, the URL-driven filter pattern	REWRITE, KEEP THE SHAPE The pattern is right; the implementation moves to the database and to shared Zod schemas.

A Technology stack

Every choice below is stated with the alternative I rejected, because several of them are genuine trade-offs rather than obvious calls.

LAYER	CHOICE	WHY THIS, OVER THE ALTERNATIVE
Language	TypeScript 5, strict	One language across web, worker and shared domain code. `strict`, `noUncheckedIndexedAccess` and `exactOptionalPropertyTypes` on from day one — retrofitting them into a 40-table codebase is miserable.
Web	Next.js 16, App Router	A marketplace lives or dies on organic search, so product, store and course pages need server rendering and stable URLs. React Server Components also ship less JavaScript to the low-end Android devices that dominate the target market. Rejected: a SPA + separate API, which doubles the deployment surface and loses SEO.
Domain logic	packages/core	Framework-free TypeScript. Next.js is transport, not the application. This is the single most important structural decision in the document — see section L.
Database	PostgreSQL 16+	Not negotiable. Real transactions, `SERIALIZABLE` isolation, CHECK and EXCLUSION constraints, deferred constraint triggers, partial indexes, `NUMERIC`, row-level locking, full-text search. The financial invariants belong in the database, not only in application code.
ORM	Drizzle ORM	See the decision block below — this one is contested.
Background jobs	pg-boss (Postgres-backed)	Lets a job be enqueued *inside the same transaction* as the ledger rows that job refers to. With Redis you get a two-phase failure: the database commits, the enqueue fails, and settlement silently never runs. Rejected: BullMQ + Redis, which is faster but buys throughput we do not have yet with a correctness hole we cannot afford. Revisit above roughly 100 jobs/second.
Auth	Better Auth + DB sessions	Phone-first OTP is a first-class requirement in this market, not an add-on. Database-backed sessions rather than stateless JWTs, because a session that controls a wallet must be revocable instantly. Rejected: Clerk and Auth0 (per-MAU USD pricing against XOF revenue), Lucia (retired upstream).
Validation	Zod	One schema per boundary — HTTP body, server action input, job payload, webhook payload — with types inferred rather than restated.
i18n	next-intl	French as the default locale, English second, locale in the URL path (`/fr/…`) so both index separately. Message catalogs live in `packages/i18n` so the worker can localise SMS and email too.
Styling	Tailwind v4 + tokens	Already in the repository, already token-driven, already dark-mode aware. No reason to change it.
Object storage	Cloudflare R2	Zero egress fees. For a platform whose core product is video in a market where bandwidth is the dominant cost, this is the difference between viable and not. S3-compatible, so the provider interface stays portable.
Video	Cloudflare Stream	Signed playback tokens, HLS adaptive bitrate, per-video access control. We store a `playback_id`, never a file URL. Alternatives worth pricing: Bunny Stream (cheaper), Mux (best tooling, most expensive). Rejected: MP4 files in a bucket — no adaptive bitrate on a 3G connection, and no way to expire access.
SMS / email	provider interface	SMS matters more than email here (OTP, order status, payout confirmations). Concrete provider chosen per country at deploy time; the interface is ours.
Observability	OpenTelemetry + Sentry	Structured logs carrying request id, actor id and ledger transaction id. A money platform without traceable financial events is not operable.

LAYER	CHOICE	WHY THIS, OVER THE ALTERNATIVE
Testing	`Vitest + Playwright`	Property-based unit tests on the ledger and commission engine; Playwright end-to-end on the purchase-to-payout path specifically.
Hosting	`containers, Postgres in eu-west-3`	Paris, not Cape Town: round-trip latency from Niamey, Bamako, Abidjan and Dakar to `eu-west-3` is materially better than to `af-south-1`. Cloudflare in front for edge caching. Vercel works for phases 0–2 but the worker is a long-running process that does not fit its model, so plan for containers.

CONTESTED DECISION

Drizzle ORM rather than Prisma

WHY

Migrations are plain SQL that a human reviews before it touches money. CHECK constraints, deferred constraint triggers and `SELECT … FOR UPDATE` are expressible directly in the schema rather than through escape hatches. No query-engine binary, which matters for cold starts and for self-hosting cheaply.

INSTEAD OF

Prisma, which has better developer experience, better relational query ergonomics and far more mature migration tooling for a 40-table schema.

TRADE-OFF

More SQL written by hand, a smaller ecosystem, and a steeper ramp for developers who have only used Prisma.

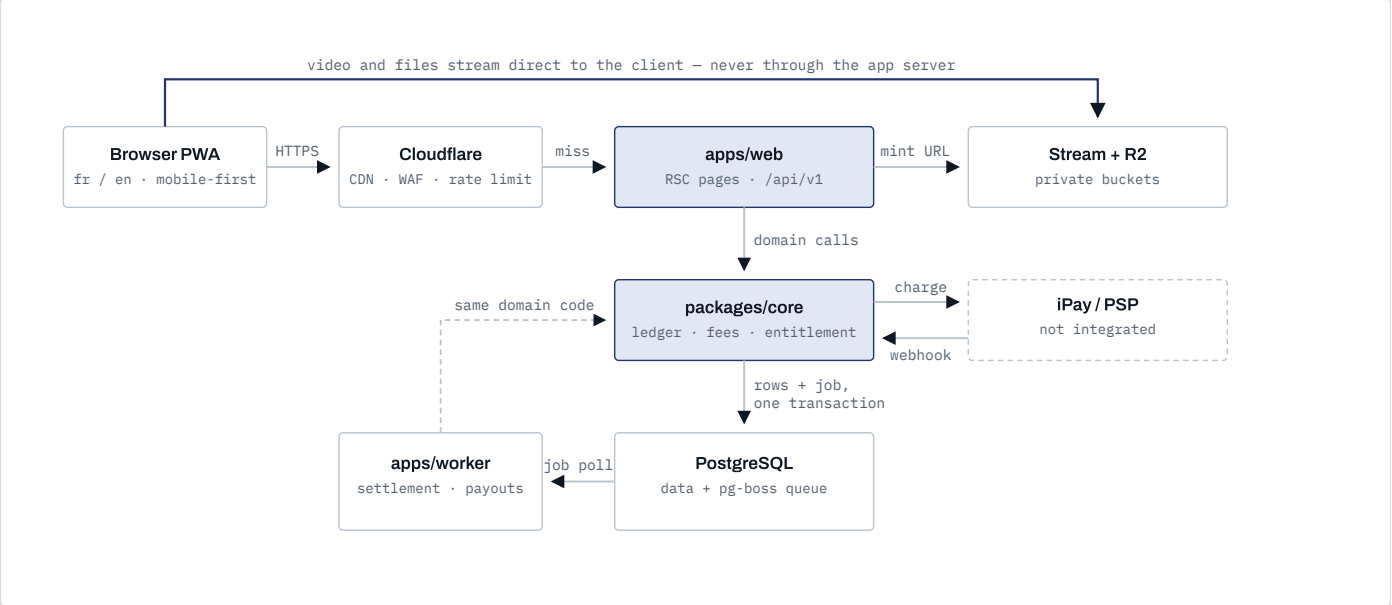
SWITCH IF

The team is junior-weighted or Prisma-experienced. In that case take Prisma and write the ledger's critical paths as raw SQL — the schema in section C is unchanged either way. Tell me the team composition and I will pick.

B System architecture

Four rules define the shape of the system. Everything else follows from them.

1. **Only `packages/core` touches the database.** Route handlers, server actions and job consumers all call the same domain services. There is no second path to the ledger.
2. **Jobs are enqueued in the transaction that creates their subject.** A settlement job and the ledger rows it will settle commit together or not at all.
3. **Media never passes through the application server.** The server mints a short-lived signed URL; the client fetches from storage directly.
4. **Webhooks are never processed inline.** Verify the signature, persist the raw event, return 200, and let the worker do the work. Payment providers retry aggressively and time out fast.



The database is the queue. Because pg-boss stores jobs in Postgres, a settlement job is inserted in the same transaction as the ledger entries it will later act on — the failure mode where money is recorded but its follow-up work is lost cannot occur. The dashed line from the worker means it imports `packages/core` rather than calling the web app.

Environments

Three: local (Docker Compose — Postgres, MinIO standing in for R2, a mock PSP), staging (real PSP sandbox, real Stream, anonymised data), production. Migrations run forward-only in CI with a tested rollback for each. Staging

must be able to run a full purchase-to-payout cycle before any production release touches the ledger.

C Database schema

Conventions that apply everywhere: UUIDv7 primary keys (time-sortable, so they index well and leak no sequence count); `timestamptz` in UTC only; status enums rather than soft deletes except where retention law requires otherwise; and **money is never a bare number** — `amount_minor bigint` always travels with a `currency char(3)`.

> **AFRICA-SPECIFIC CORRECTNESS TRAP**
>
> XOF and XAF have **zero** decimal places. NGN (kobo) and GHS (pesewa) have two. Any code that assumes "minor units means cents, divide by 100" will be wrong by a factor of 100 across the entire UEMOA zone. The exponent comes from the `currencies` table, never from a constant.

The financial core

This is the part I would write first and change least. The rest of the schema can evolve; this cannot.

```sql
-- Money is always a pair, and the exponent is data, not a constant.
CREATE TABLE currencies (
  code        char(3)  PRIMARY KEY,          -- XOF, XAF, NGN, GHS
  minor_unit  smallint NOT NULL CHECK (minor_unit BETWEEN 0 AND 4),
  is_active   boolean  NOT NULL DEFAULT true
);

CREATE TYPE ledger_account_kind AS ENUM (
  'user_available',  -- withdrawable
  'user_pending',    -- earned, not yet releasable
  'platform_escrow', -- captured, owed to somebody, not yet split
  'platform_revenue',
  'psp_clearing',    -- money at the payment provider
  'payout_payable'   -- approved withdrawal, not yet paid out
);

CREATE TABLE ledger_accounts (
  id          uuid PRIMARY KEY,
  kind        ledger_account_kind NOT NULL,
  owner_type  text,                        -- 'user' | 'store' | NULL for platform
  owner_id    uuid,
  currency    char(3) NOT NULL REFERENCES currencies(code),
  UNIQUE (kind, owner_type, owner_id, currency)
);

CREATE TABLE ledger_transactions (
  id               uuid PRIMARY KEY,          -- uuidv7
  kind             text NOT NULL,              -- capture | settlement | refund | payout …
  occurred_at      timestamptz NOT NULL DEFAULT now(),
  idempotency_key  text UNIQUE,
  reverses_id      uuid REFERENCES ledger_transactions(id),
  metadata         jsonb NOT NULL DEFAULT '{}'   -- order_id, payment_id, actor…
);

CREATE TABLE ledger_entries (
  id              uuid PRIMARY KEY,
  transaction_id  uuid    NOT NULL REFERENCES ledger_transactions(id),
  account_id      uuid    NOT NULL REFERENCES ledger_accounts(id),
  direction       smallint NOT NULL CHECK (direction IN (-1, 1)),
  amount_minor    bigint   NOT NULL CHECK (amount_minor > 0),
  currency        char(3)  NOT NULL REFERENCES currencies(code)
);

-- Append-only. The application role cannot rewrite history.
REVOKE UPDATE, DELETE ON ledger_entries, ledger_transactions FROM afrinext_app;

-- Every transaction balances, per currency. Deferred, so all entries
-- can be inserted before the invariant is checked at COMMIT.
CREATE CONSTRAINT TRIGGER ledger_must_balance
  AFTER INSERT ON ledger_entries
  DEFERRABLE INITIALLY DEFERRED
  FOR EACH ROW EXECUTE FUNCTION assert_transaction_balanced();

-- A cache, never the truth. Rebuilt and asserted nightly.
CREATE TABLE account_balances (
  account_id    uuid PRIMARY KEY REFERENCES ledger_accounts(id),
  balance_minor bigint NOT NULL DEFAULT 0,
  entry_count   bigint NOT NULL DEFAULT 0,
  updated_at    timestamptz NOT NULL DEFAULT now()
);
```

Entities by module

The brief listed candidate tables and asked me not to create them blindly. This is the filtered set, with the ones I would *not* build in phase 1 marked.

MODULE	TABLES	NOTES
Identity	`users`, `user_identities`, `sessions`, `profiles`, `roles`, `permissions`, `role_permissions`, `role_assignments`, `countries`, `currencies`, `locales`	`user_identities` holds one row per credential (phone, email, future OAuth) so a user can add a second login method without a schema change. `role_assignments` is scoped — see section D.
Stores	`stores`, `store_settings`, `store_payout_accounts`, `STORE_MEMBERS`	The *store* is the seller, not the user — orders reference `store_id`. `store_members` is deferred until multi-person stores are actually requested.
Catalog	`categories`, `products`, `product_media`, `digital_assets`, `PRODUCT_VARIANTS`, `INVENTORY_ITEMS`, `SERVICE_OFFERINGS`	One `products` table with a `kind` discriminator plus kind-specific side tables. Variants and inventory arrive with physical goods; service offerings with the services module.
Learn	`courses`, `course_modules`, `course_lessons`, `lesson_assets`, `course_enrollments`, `lesson_progress`, `COURSE_CERTIFICATES`	Certificates are a future feature in the brief and stay that way — but `enrollments` carries the completion data a certificate would need.
Commerce	`carts`, `cart_items`, `checkout_groups`, `orders`, `order_items`, `order_fee_lines`, `order_events`	`order_fee_lines` is the frozen commission snapshot — the reason changing a commission rate never rewrites history.
Payments	`payment_intents`, `payment_attempts`, `payment_events`, `refunds`	`payment_events` stores the raw provider payload with a `UNIQUE (provider, provider_event_id)` — this single constraint is what makes webhook replay harmless.
Ledger	`ledger_accounts`, `ledger_transactions`, `ledger_entries`, `account_balances`, `settlement_holds`, `commission_rules`, `idempotency_keys`	Above.
Payouts	`withdrawal_requests`, `payout_batches`, `kyc_records`	KYC tiers gate withdrawal ceilings. See section K.
Delivery	`delivery_partners`, `delivery_orders`, `delivery_events`, `delivery_proofs`, `CASH_COLLECTIONS`	Deferred to the physical-goods phase. `cash_collections` models the rider's cash float — see section I.
Referral	`referral_programs`, `referral_codes`, `referral_attributions`, `referral_commissions`	No `referral_relationships` tree — single level only, see section H.
Trust & ops	`reviews`, `notifications`, `audit_logs`, `platform_settings`, `feature_flags`, `DISPUTES`, `DISPUTE_MESSAGES`	Disputes arrive with physical goods, where they actually occur. Digital refunds in phase 1 are an admin action against an audited reversal.

Tables from the brief I would **not** create: `referral_relationships` (implied by single-level attribution), `user_roles` (superseded by scoped

`role_assignments`), and a separate `services` table (services are `products` with `kind = 'service'`).

D Authentication & permissions

Authentication

- **Phone + OTP is the primary method**, email + password secondary. Phone penetration far exceeds email use in the target countries, and phone is already the identifier for mobile money.
- **Database sessions, not JWTs.** An opaque token in an `httpOnly; Secure; SameSite=Lax` cookie, resolved against `sessions` on each request. A JWT cannot be revoked before it expires; a session that can move money must be killable in one query.
- Session rotation on privilege change, sign-in and password change. Shorter idle timeout for admin sessions than for shoppers.
- **Step-up re-verification** for money-sensitive actions: requesting a withdrawal, adding or changing a payout account, changing the phone number.

Authorization

Roles are scoped, additive rows. A single user simultaneously being a buyer, an instructor, a rider in Niamey and staff on someone else's store is the normal case, not an edge case.

```
CREATE TABLE role_assignments (
  id          uuid PRIMARY KEY,
  user_id     uuid NOT NULL REFERENCES users(id),
  role_id     uuid NOT NULL REFERENCES roles(id),
  scope_type  text NOT NULL,   -- 'global' | 'store' | 'course' | 'zone'
  scope_id    uuid,             -- NULL when scope_type = 'global'
  granted_by  uuid REFERENCES users(id),
  granted_at  timestamptz NOT NULL DEFAULT now(),
  UNIQUE (user_id, role_id, scope_type, scope_id)
);
```

Permissions are strings of the form `resource.action` — `product.publish`, `order.refund`, `withdrawal.approve`, `lesson.view`. Roles hold sets of them. Every check goes through one function:

```
// packages/core/authz — the only permission gate in the system.
await authorize(actor, 'product.publish', { type: 'store', id: storeId });
```

RULE

IDOR is prevented by **scoping the query**, not by fetching a row and then checking it. Repository methods take the actor and constrain in SQL: a seller's order list is `WHERE store_id = ANY(actor.storeIds)`, never `findById()` followed by an ownership test. The second pattern works until someone forgets it once.

ADMINISTRATIVE ROLES ARE SPLIT BY LEAST PRIVILEGE

ROLE	CAN	CANNOT
`support`	Read users, orders, enrollments; resend receipts; open disputes	Touch the ledger, approve payouts, change commission rules
`ops`	Moderate catalog, verify stores, manage categories and delivery zones	Anything financial
`finance`	Approve withdrawals, issue refunds, post adjustments, view the ledger	Edit catalog, change permissions, alter user identities
`superadmin`	Role and settings administration	Approve their own withdrawal — a hard, code-level self-approval block

E Financial architecture

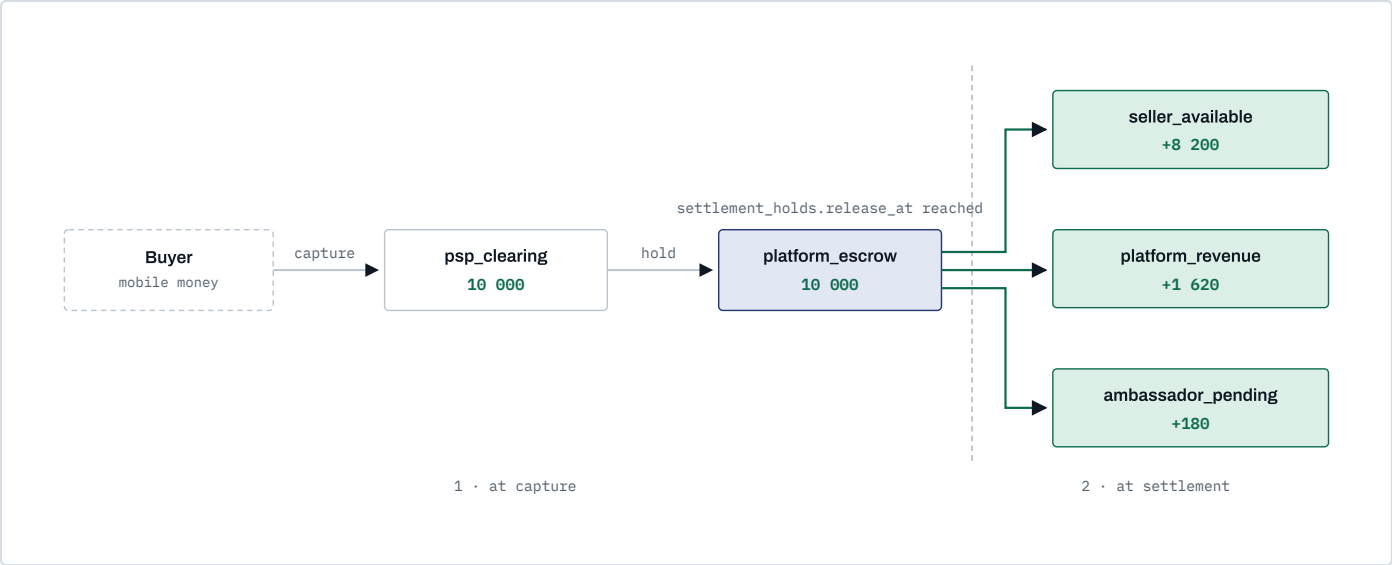
The wallet is not a balance field with rules around it. It is a **double-entry ledger**, and the balance is a derived quantity that we cache for speed and verify continuously.

The four invariants

1. **Entries are immutable.** No UPDATE, no DELETE — enforced by revoked grants, not by convention. A mistake is corrected by posting a compensating transaction linked through `reverses_id`.
2. **Every transaction balances.** Per transaction, per currency, the signed sum of entries is zero. A deferred constraint trigger rejects the COMMIT otherwise. Money cannot be created or destroyed by application code.
3. **Balances are cached, not authoritative.** `account_balances` is updated inside the same transaction under a row lock; a nightly job recomputes `SUM(direction × amount_minor)` per account and alarms on any drift.
4. **No floating point, anywhere.** `bigint` minor units and a currency code, from the database through the API to the rendered price.

Pending versus available

These are not statuses on a row. They are **two different accounts**, and "funds becoming withdrawable" is a transfer between them. That means the question "why is my money still pending?" is answerable by pointing at a specific ledger transaction and its `settlement_holds` row, rather than by reading application logic.



One 10 000 XOF course sale, at an 18% platform commission with a 10% referral program. Every arrow is one balanced ledger transaction. Note where the referral is drawn from: 180 is 10% of the *platform's* 1 800, not of the gross — so referral cost can never exceed what the platform earned on that sale. The ambassador's share lands in `pending` and is released separately once the buyer's refund window closes. For physical goods the seller's share also lands in `seller_pending` and moves to `available` on a second transaction when the delivery hold expires.

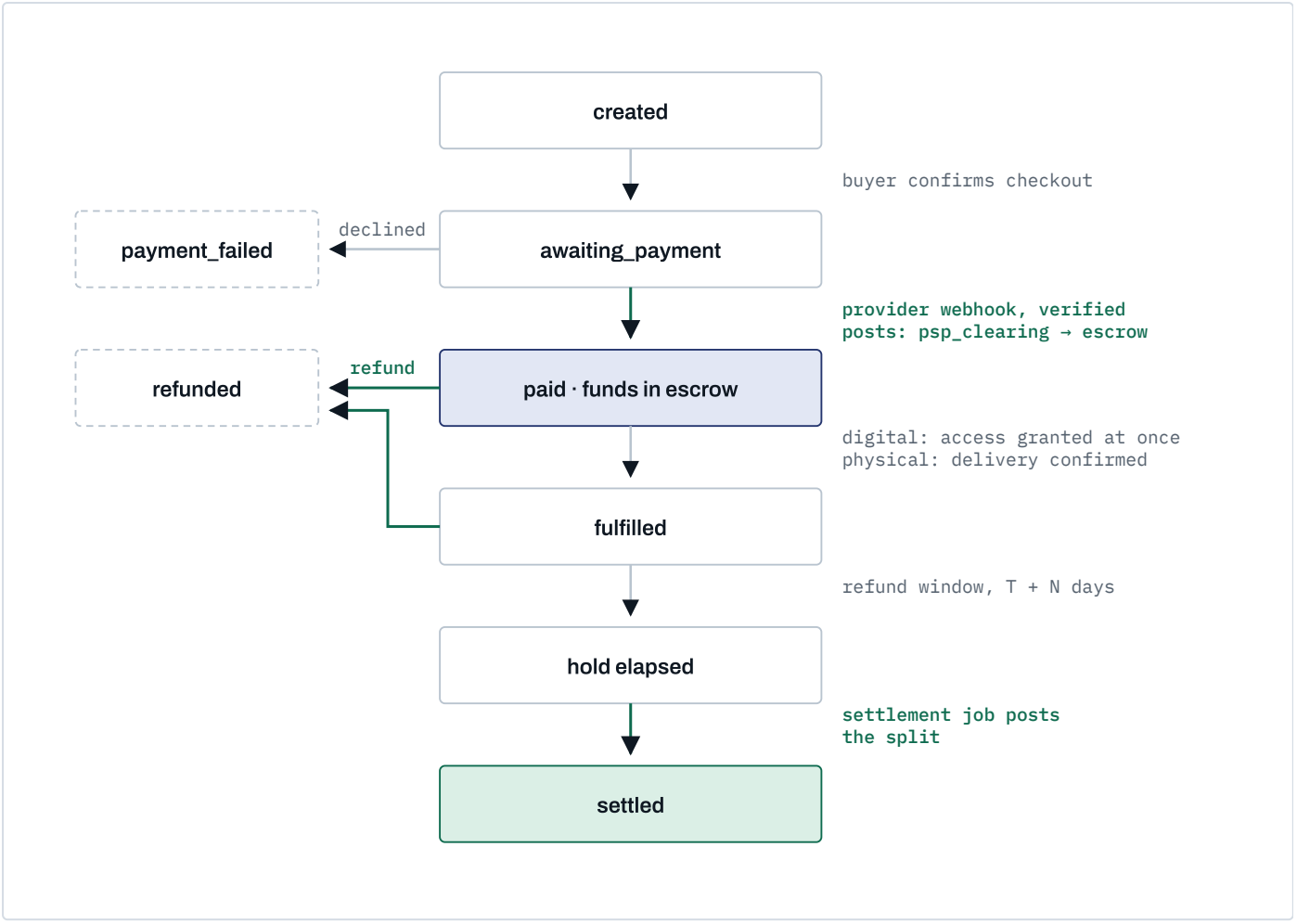
Rounding: the residual rule

Commission percentages rarely divide cleanly into whole francs. The rule is fixed and applied everywhere:

1. Compute every non-seller line (platform, referral, delivery) in minor units, rounding **half up**.
2. The seller's line is `gross − Σ(all other lines)`, never computed independently.

The split therefore re-sums to the gross by construction. No minor unit is ever created or lost to rounding, and the residual consistently favours the seller — a defensible position if it is ever questioned.

When money moves



Green transitions post ledger entries; grey ones only change order state. Money enters escrow when the provider webhook is verified — never when the buyer returns from a redirect. A dispute does not get its own state: it *extends* the hold by pushing `settlement_holds.release_at` forward, so disputed money simply never reaches `available` while the dispute is open.

The commission engine

Rules are data. Resolution happens once, at order creation, and the result is frozen.

```
CREATE TABLE commission_rules (
  id              uuid PRIMARY KEY,
  transaction_type text NOT NULL,   -- digital | course | physical | service | delivery | referral
  country_code    char(2),              -- NULL = any
  category_id     uuid,                 -- NULL = any
  store_id        uuid,                 -- NULL = any; a negotiated rate for one seller
  rate_bps        integer,              -- basis points; 1800 = 18.00%
  fixed_minor     bigint,
  currency        char(3) REFERENCES currencies(code),
  priority        integer NOT NULL,
  effective_from  timestamptz NOT NULL,
  effective_to    timestamptz
);
```

Resolution is most-specific-wins, broken by `priority` and then by `effective_from` — deterministic, and unit-testable in isolation. The resolved outcome is written to `order_fee_lines` with the `rule_id`, the basis and the computed amount.

> **WHY THE SNAPSHOT MATTERS**
>
> Raise the platform commission from 18% to 20% next quarter and **no historical order changes.** Every past settlement can still be re-derived from its own fee lines, and any seller asking "why did I receive this amount?" gets an answer that references the rule that was live at the moment they sold. A system that recomputes fees from current rules cannot answer that question truthfully.

Withdrawals

1. Request → validated against KYC tier ceiling, minimum amount, available balance, payout-account cooling period, and absence of open disputes.
2. Approved (automatically under a configured threshold, by `finance` above it) → ledger posts `user_available → payout_payable`. The money leaves the withdrawable balance at approval, so it cannot be double-spent while the payout is in flight.
3. Payout executed through the provider → on confirmation, `payout_payable → psp_clearing`.
4. On failure, a compensating transaction returns the amount to `user_available` and the request is marked failed with the provider's reason.

F Marketplace architecture

One products table, three kinds

`products.kind ∈ {physical, digital, service}`, with kind-specific side tables (`product_variants` + `inventory_items`, `digital_assets`, `service_offerings`). Not three separate tables — because carts, order items, reviews, search, favourites and fee resolution all need a single polymorphic target, and a three-table design forces a nullable triple foreign key into `order_items` that every query then has to unpack.

What differs between kinds is **fulfilment and hold policy**, and that lives in one place:

KIND	FULFILLED WHEN	DEFAULT HOLD
digital	Payment verified	24 h
course	Enrollment created	72 h
service	Buyer confirms, or auto after N days	7 d
physical	Delivery confirmed with proof	7 d

These are defaults in `platform_settings`, overridable per country and per category from the admin console — not constants in code.

Stores are the seller

Orders reference `store_id`, never `user_id`. This costs nothing now and buys three things later: a user can run more than one store; a store can gain staff without reshaping the order history; and a store can be transferred or suspended without touching the person's account, wallet or other roles.

Multi-store carts

A buyer will put a course from Dakar and a bag of shea butter from Bamako in one cart. At checkout the cart splits into **one order per store**, grouped under a `checkout_group_id`, paid by a single payment intent. Each order then settles independently on its own timeline — the course in 72 hours, the shea butter when it is delivered. Attempting to keep them as one order forces every fulfilment and refund decision to be all-or-nothing.

Search

Postgres full-text search with a generated `tsvector` column, plus `pg_trgm` for typo tolerance, and a French text-search configuration alongside English. Facets come from ordinary indexed columns. Move to Meilisearch or Typesense when the catalog passes roughly 100 000 rows or when faceted search gets heavy — not before, and not to Elasticsearch, which is an operational burden this team should not carry at launch.

Reviews

Only from a verified purchase: `reviews.order_item_id` is a required foreign key, unique per buyer per item. This kills fake-review farming structurally rather than through moderation, which matters when reviews feed seller ranking and seller ranking feeds income.

G Afrinext Learn & video access

Structure is conventional — `courses → course_modules → course_lessons → lesson_assets`, with lesson kinds `video | pdf | text`. The interesting parts are entitlement and bandwidth.

Entitlement is checked at token-mint time, on every request

1. The player asks `GET /api/v1/learn/lessons/:id/playback`.
2. The server resolves the actor, checks `course_enrollments` for a live entitlement (and `access_expires_at` if the course is time-limited), and calls `authorize(actor, 'lesson.view', {course})`.
3. Only then does it mint a **signed Stream token, scoped to that user, valid 2–5 minutes**, and return an HLS manifest URL.
4. The player refreshes the token mid-playback as it nears expiry.
5. Every mint is written to `audit_logs` — which also gives instructors real "who watched what" analytics for free.

> **NEVER**
>
> Paid course content never has a permanent public URL. Not in a public bucket, not behind an unguessable path, not with a long-lived signed URL. An unguessable URL is a password that gets forwarded on WhatsApp.

Downloads are a per-asset instructor choice

`lesson_assets.downloadable boolean`, set by the instructor per asset. When true, downloading issues a separate short-TTL signed URL, logged, and rate-limited per user. When false, no download URL is ever minted — the flag is enforced server-side at mint time, not by hiding a button.

What is realistic about protection

> **BE HONEST WITH INSTRUCTORS**
>
> Signed short-lived URLs plus a burned-in overlay carrying the viewer's phone number make casual re-sharing traceable and inconvenient. They do **not** prevent a determined screen recording. Genuine prevention requires Widevine/FairPlay DRM, which is expensive and out of scope here. Instructors should be told this plainly rather than sold an assurance the platform cannot honour.

Progress and resume

`lesson_progress(enrollment_id, lesson_id, position_seconds, furthest_seconds, completed_at)`, written by a heartbeat every ~15 seconds, throttled, and flushed on pause and on unload via `sendBeacon`. `course_enrollments.last_lesson_id` gives cross-device resume: the learner picks up on their phone exactly where they stopped on a shared laptop. Keeping `furthest_seconds` separate from `position_seconds` means scrubbing backwards does not destroy completion state.

Built for the actual network

- An encoding ladder that **starts at 240p**, not at 720p. Adaptive bitrate matters far more here than peak quality.
- An audio-only rendition for lecture-style courses, selectable by the learner to save data.
- Progress writes are a few dozen bytes and batched — a learner should never lose their place because the heartbeat could not get through.
- Where the instructor permits downloads, offline viewing is the difference between a finished course and an abandoned one.

H Afrinext Network (referrals)

The brief's principle — pay for real economic activity, never for recruitment — is enforced structurally here, not by policy text.

Attribution

- An ambassador gets a code; the link is `/r/:code`, which sets a first-party cookie (90 days) and redirects.
- At signup the cookie is consumed into `referral_attributions(referrer_user_id, referred_user_id, program_id, attributed_at, expires_at)`, with `UNIQUE(referred_user_id)`.
- **First touch, immutable.** The first ambassador to introduce someone keeps the attribution. Last-touch invites a scramble to overwrite each other's referrals just before a purchase.

Qualification

A commission is created only when a **settled ledger transaction** of a configured type exists — not at signup, not on profile completion, not on "becoming active". There is no code path that pays for a registration, because commissions are computed by a job that reads settled transactions and has no other input.

CONFIGURABLE PER PROGRAM	EXAMPLE
Basis	`platform_revenue`
Rate	`1000 bps (10%)`
Duration from attribution	`365 days`
Per-referral lifetime cap	`150 000 XOF`
Eligible transaction types	`course, digital, physical, service`
Country scope	`NE, ML, BF …`

RECOMMENDATION, NEEDS YOUR SIGN-OFF

Single level only — no downline of a downline

WHY	Multi-level structures are where recruitment-scheme regulatory attention concentrates, and they contradict the brief's own stated principle. One level keeps the model explainable to a regulator in one sentence: "we share part of our commission with whoever introduced the seller."
INSTEAD OF	Two or more tiers, which measurably increases ambassador enthusiasm and measurably increases legal exposure.
TRADE-OFF	Slower ambassador-network growth than a tiered scheme would produce.
NOTE	Because the commission basis is platform revenue, the payout is bounded by definition — the platform can never owe more in referral than it earned on the sale. That guardrail is what makes the program safe to run at all, and it disappears if the basis is ever changed to gross.

Fraud controls

- Self-referral detection across phone number, device fingerprint, payout account and payment instrument.
- Referral commissions land in `ambassador_pending` and release only after the underlying order clears its refund window — so a wash sale that is refunded never pays out.
- Velocity limits per ambassador, with automatic hold and review above configured thresholds.
- Circular attribution (A refers B refers A) rejected at write time.

I Delivery architecture

Deferred to the physical-goods phase, but designed now because it constrains the order state machine.

STATE MACHINE

`requested → offered → accepted → picked_up → in_transit → delivered`, with `failed` and `cancelled` as terminal branches. Every transition writes a `delivery_events` row with actor, timestamp and GPS point — the audit trail is the record, the current status is a cached projection of it.

DISPATCH

Offers broadcast to eligible partners in the zone; first acceptance wins via a conditional update (`UPDATE … WHERE status = 'offered'`) so two riders can never both hold the same job. Unaccepted offers expire and re-broadcast with a widening radius.

PROOF OF DELIVERY

A short OTP shown in the buyer's app and entered by the rider, plus an optional photo and the GPS fix at completion. The OTP is what flips the order to `fulfilled` and starts the settlement hold — a rider cannot self-declare a delivery into existence.

RIDERS LOSE SIGNAL CONSTANTLY

The rider interface queues state transitions locally and replays them with idempotency keys when connectivity returns. A transition applied twice must be a no-op; a transition applied out of order must be rejected by the state machine, not silently accepted.

> **CASH ON DELIVERY IS A BIGGER PROBLEM THAN IT LOOKS**
>
> COD is the dominant payment method for physical goods across much of the region, and it inverts the money flow: the rider collects cash, so the platform is owed money by the rider rather than holding the buyer's funds. That needs a `cash_collections` liability account per rider, a deposit and reconciliation process, float limits, and a real operational routine — it is a project, not a checkbox. **My recommendation is to launch physical goods prepaid-only** and add COD as its own phase with its own design.

J API architecture

Two entry styles, one implementation. **Server Actions** for first-party form mutations (fewer round trips, progressive enhancement) and **versioned REST** at `/api/v1/*` for everything a future native app, an ambassador tool or a partner integration would consume. Both call the same `packages/core` services. Logic is never duplicated between them, and neither one is allowed a private path to the database.

DOMAIN	REPRESENTATIVE ENDPOINTS
auth	POST /auth/otp/request · /auth/otp/verify · /auth/session · DELETE /auth/session
me	GET /me · PATCH /me · GET /me/roles · GET /me/orders · GET /me/enrollments
stores	POST /stores · GET /stores/:slug · PATCH /stores/:id · GET /stores/:id/analytics
catalog	GET /products · GET /products/:slug · POST /stores/:id/products · POST /products/:id/publish
search	GET /search?q&kind&category&country&cursor
cart · checkout	GET/POST /cart/items · POST /checkout · GET /checkout/:groupId
orders	GET /orders · GET /orders/:id · POST /orders/:id/cancel · POST /orders/:id/refund
payments	POST /payments/intents · GET /payments/:id · POST /webhooks/:provider
wallet	GET /wallet · GET /wallet/transactions · POST /wallet/withdrawals · GET /wallet/withdrawals/:id
learn	GET /courses/:slug · GET /learn/lessons/:id · POST /learn/lessons/:id/progress · GET /learn/lessons/:id/playback
delivery	GET /delivery/offers · POST /delivery/:id/accept · POST /delivery/:id/transition
referrals	GET /referrals/me · GET /referrals/commissions · GET /r/:code
admin	/admin/* — mirrors the above under `admin.*` permissions

CONVENTIONS

- `Idempotency-Key` **is required** on every POST that moves money. The key, the request hash and the resulting transaction id are stored; a replay returns the original response rather than acting twice.
- Cursor pagination everywhere (`?cursor=&limit=`). Offset pagination on a growing catalog silently skips and duplicates rows.
- Responses are `{ data, meta }`; errors are `{ error: { code, message, fields } }` with **stable machine-readable codes** — clients must never branch on message text, which is localised.
- Money crosses the wire as `{ amountMinor: 8200, currency: "XOF" }`. Never a formatted string, never a float.
- Rate limits by route class: strictest on OTP send and verify, then auth, then write, then read.
- Webhooks return `200` as fast as possible after signature verification and persistence. All work happens in the worker.

K Security architecture

CONTROL	WHERE IT IS ENFORCED
Authentication	Opaque session tokens in `httpOnly; Secure; SameSite=Lax` cookies, resolved server-side. Rotated on sign-in and privilege change. Argon2id for the password path.
Step-up verification	Fresh OTP required for withdrawal requests, payout-account changes and phone changes — regardless of session age.
Authorization	One `authorize()` choke point in `packages/core/authz`. No route handler makes its own decision.
IDOR	Prevented by scoping queries to the actor in SQL, not by post-fetch ownership checks.
Input validation	Zod at every boundary; database CHECK and FK constraints as the last line. Frontend validation is a convenience and is never trusted.
Rate limiting	Per IP and per user, at the edge and in the app. Hardest limits on OTP send/verify, login, withdrawal, and playback-token minting.
Webhook integrity	Signature verified against the **raw** body before parsing; timestamp freshness window; `UNIQUE (provider, provider_event_id)` makes replay a no-op.
File access	Private buckets only. Short-TTL signed URLs minted after an entitlement check. Every mint audited.
Payment card data	Never touched. Provider-hosted collection only, which keeps PCI scope at SAQ-A. This is a deliberate architectural constraint, not an oversight.
Audit	Append-only `audit_logs` for every admin action and every financial state change: actor, action, target, before/after, IP, user agent, request id.
Fraud	Velocity checks, device fingerprinting, a cooling period before first withdrawal to a newly added payout account, and automatic hold-and-review thresholds.
Separation of duty	No user can approve a withdrawal to an account they control. Enforced in code, not in policy.
Secrets	Environment only, distinct per environment, rotated. Sandbox and production provider credentials never coexist in one config.
Data protection	PII minimised and encrypted at rest; retention policy per data class; export and deletion paths built for Nigeria's NDPR, Ghana's Data Protection Act and UEMOA national regimes.

EXPLICITLY OUT OF SCOPE, AND NAMED AS SUCH

DRM for video (Widevine/FairPlay). A third-party penetration test — which should be scheduled and budgeted *before* public launch, not after. Formal PCI-DSS certification beyond SAQ-A. Automated AML transaction-monitoring beyond threshold rules.

L Project structure

pnpm workspaces plus Turborepo. The directory layout is not cosmetic — it is what physically prevents domain logic from leaking into the framework.

```
afrinext/
├─ apps/
│  ├─ web/                 Next.js 16 — pages, route handlers, server actions
│  └─ worker/              long-running pg-boss consumers
├─ packages/
│  ├─ core/                framework-free domain logic — the application
│  │  ├─ money/            Money type, minor units, rounding, formatting
│  │  ├─ ledger/           posting, balances, reconciliation
│  │  ├─ commissions/      rule resolution, fee snapshots
│  │  ├─ orders/           order + settlement state machines
│  │  ├─ payments/         PaymentProvider interface + registry
│  │  ├─ learn/            entitlement, playback tokens, progress
│  │  ├─ delivery/
│  │  ├─ referrals/
│  │  └─ authz/            permissions, authorize()
│  ├─ db/                  Drizzle schema, SQL migrations, seeds
│  ├─ providers/           psp-ipay, psp-manual, storage-r2, video-stream, sms, email
│  ├─ ui/                  design system (seeded from today's prototype)
│  ├─ i18n/                fr / en message catalogs, shared with the worker
│  └─ config/              tsconfig, eslint, tailwind preset
├─ docs/
│  ├─ architecture/        this document
│  ├─ adr/                 architecture decision records, numbered
│  └─ runbooks/            reconciliation drift, stuck payout, failed webhook
└─ infra/                  docker, compose, CI, deploy
```

THE RULE THAT MAKES THIS WORTH DOING

`packages/core` may import `packages/db`. It may **not** import anything from `next`, and this is enforced by an ESLint boundary rule in CI, not by discipline. The consequence: the ledger and commission engine are testable in milliseconds without a server, the worker runs identical business logic to the web app, and extracting a standalone API service later is a packaging change rather than a rewrite.

M Development roadmap

Phases are ordered by *irreversibility*, not by visibility. The money spine comes before any product feature, because it is the thing that cannot be retrofitted. Durations assume two to three engineers and are estimates, not commitments.

PHASE	SCOPE	EXIT CRITERION — THE THING THAT MUST DEMONSTRABLY WORK	EST.
P0 Foundations	Monorepo, Postgres, Drizzle schema v1, auth (phone OTP + email), scoped RBAC, i18n, design system extraction, CI, observability skeleton	A user registers by phone, signs in, switches language. CI runs typecheck, lint, tests and migrations forward and back on every commit.	2 wk
P1 Money spine	`money`, `ledger`, commission engine, idempotency, settlement holds, reconciliation job, admin ledger viewer	Property-based tests over 10 000 randomised operations prove every transaction balances and the cached balance equals the sum of entries. An injected drift is caught by the reconciliation job. **No product feature is built before this passes.**	3 wk
P2 Stores & catalog	Store creation, digital products and courses as products, categories, media upload to R2, public storefront and product pages, search	A seller publishes a paid digital product that renders server-side at a public URL with correct metadata and appears in search.	3 wk
P3 Cycle closed, no PSP	Cart, checkout, orders, fee snapshots, and a `ManualTransferProvider` where an admin confirms receipt	The complete economic cycle runs end to end without any payment integration: buyer orders → admin confirms → ledger settles → seller sees earnings → requests withdrawal → finance marks it paid. **This de-risks everything that depends on iPay.**	3 wk
P4 iPay	Real provider implementation behind the existing interface	In sandbox: charge created, webhook signature verified, status query reconciled, refund executed, replay proven harmless. GATED Cannot start until the documentation and sandbox credentials in section P are in hand.	unknown
P5 Afrinext Learn	Modules, lessons, Stream upload, entitlement, signed playback, progress and resume, download flag, learner interface	An automated test proves a paid lesson is unplayable without entitlement, and that resume works across two devices for one learner.	4 wk
P6 Wallet & payouts	Wallet interface, KYC tiers, withdrawal flow with step-up auth, payout batches, finance console	Full withdrawal lifecycle including a provider failure that correctly reverses back to available balance.	3 wk
P7 Network	Referral programs, attribution, qualification, commissions, ambassador dashboard, fraud checks	Tests prove a commission can only originate from a settled transaction, that self-referral is blocked, and that a refunded order pays nothing.	3 wk
P8 Physical & delivery	Variants, inventory, shipping, delivery partner interface, state machine, proof of delivery, delivery earnings	A physical order settles only on OTP-confirmed delivery, and a rider replaying a queued transition twice changes nothing.	5 wk
P9 Services & creators	Service offerings, booking and request flow, creator categories and portfolios	A service settles on buyer confirmation, and on auto-confirmation after the configured window.	3 wk
P10 Hardening	Load test, penetration test, runbooks, restore drill, compliance review, admin settings completeness	A database restore is performed from backup against a stopwatch, and the recovery time is written down.	3 wk

The admin console is **not** a phase. Each phase ships the admin surface for what it built; an admin console deferred to the end is an admin console that never gets finished.

To the recommended launch vertical (P0–P3, P5, P6, plus P4 whenever iPay unblocks): roughly **18 weeks**. Full scope as briefed: roughly **32 weeks**. If iPay stalls, P3's manual rail means you can onboard real sellers and take real money by bank transfer while it is resolved.

R Risk register

RISK 1 — THE ONE THAT CAN STOP THE PROJECT

Holding customer funds may be a regulated activity. In the UEMOA zone, taking money from a buyer, holding a seller's balance and disbursing it can fall under BCEAO e-money and payment-services rules; Nigeria and Ghana have their own regimes. This is not an engineering problem and I cannot solve it in the architecture — but the architecture can be shaped to reduce exposure by keeping funds at the licensed provider and having Afrinext record *entitlement* rather than custody. **Get local counsel before P6 (payouts) ships.** Please tell me who is advising on this.

RISK	SEV	MITIGATION
Ledger correctness — a bug here is money, and it is discovered by users	HIGH	Double-entry with database-enforced balancing, immutability by revoked grants, property-based tests, nightly reconciliation with alarms, ledger built in P1 before anything depends on it
iPay is an unknown — the API may lack payouts or refunds entirely	HIGH	Provider abstraction plus the manual rail in P3, so no other phase is blocked. Section P lists exactly what is needed to remove this risk
Referral fraud and scheme perception	MED	Single level, commission drawn from platform revenue only, qualification from settled transactions only, pending-until-refund-window, self-referral detection
Scope — the brief is a 32-week product and enthusiasm compounds	HIGH	The launch vertical in section 00; phases with hard exit criteria; nothing starts until the prior phase's criterion demonstrably passes
KYC and AML on withdrawals	MED	Tiered KYC with withdrawal ceilings per tier; identity verification before the first payout; threshold-based holds. Provider choice still open
Mobile money reversal semantics differ from cards and vary by operator	MED	Conservative hold windows, configurable per country and per provider; never assume card-like chargeback behaviour
Video bandwidth cost and learner data cost	MED	R2 zero-egress, ladder starting at 240p, audio-only rendition, permitted offline download
Latency from West Africa	LOW	Postgres in `eu-west-3`, Cloudflare edge caching. Measure real round-trip from Niamey, Lagos and Accra in P0 and revisit if the numbers disagree with the assumption
Cash on delivery reconciliation	MED	Recommend launching physical goods prepaid-only; COD gets its own phase and its own liability model
Cross-jurisdiction data protection	MED	PII minimisation, encryption at rest, per-class retention, export and deletion paths built in P0 rather than bolted on
Operating a money system on-call	MED	Runbooks written in the phase that creates the failure mode; reconciliation alarms route to a human; restore drill rehearsed in P10

P What the iPay integration needs

> **NOTHING IS INTEGRATED**
>
> No iPay endpoint, parameter, signature scheme or webhook payload appears anywhere in this document, and none will be written until the official documentation and sandbox credentials are in hand. What exists in the plan is an *interface* that iPay will be implemented behind. Any code claiming otherwise would be fiction.

The abstraction Afrinext codes against:

```
interface PaymentProvider {
  createCharge(input: ChargeInput): Promise<ChargeResult>;
  getCharge(providerRef: string): Promise<ChargeStatus>;
  verifyWebhook(rawBody: Buffer, headers: Headers): Promise<VerifiedEvent>;
  refund(providerRef: string, amount: Money): Promise<RefundResult>;
  createPayout?(input: PayoutInput): Promise<PayoutResult>;   // optional — may not exist
  getPayout?(providerRef: string): Promise<PayoutStatus>;
}
```

`createPayout` is optional on purpose. If iPay has no disbursement API, payouts run as an operational batch and the platform still works — that fact should be established early, because it changes the finance team's workload substantially.

REQUIRED FROM THE OFFICIAL IPAY DOCUMENTATION, BEFORE P4 CAN BEGIN

1. Base URLs for sandbox and production.
2. Authentication scheme — API key, HMAC, OAuth client credentials — and where credentials belong in a request.
3. Charge initiation: request and response schema, and the supported channels per country (Orange Money, MTN MoMo, Moov, Wave, Airtel, card, bank).
4. Flow type: hosted redirect, inline widget, or USSD push. This changes the checkout UI, not just the backend.
5. Status query endpoint and the complete status enumeration, including every terminal and non-terminal value.
6. Webhooks: how the URL is registered, payload schema, signature algorithm and header name, timestamp tolerance, and retry policy.
7. Currency handling and the minor-unit convention — specifically whether XOF amounts are sent as whole francs.
8. Whether a disbursement or payout API exists, and its settlement timing.
9. Refund API and its constraints — partial refunds, time limits.

10. Idempotency support on charge creation, and rate limits.
11. Settlement schedule to the merchant account (T+N), and the fee structure.
12. Sandbox credentials plus test phone numbers or accounts for each channel.

Items 6, 7 and 8 are the ones that most often invalidate an assumed design. If only a subset is available now, send items 1–6 and P4 can begin against a partial implementation.

S Status register

The brief asked that anything unimplemented be marked as such. This is that list, and it is complete as of this document.

ITEM	STATUS
Everything in this blueprint	`NOT IMPLEMENTED` Architecture only. No code from this plan has been written.
iPay payment integration	`NOT IMPLEMENTED` Blocked on the documentation in section P. Not designed against, not tested, not stubbed with invented shapes.
Database, authentication, wallet, ledger, courses, delivery, referrals	`NOT IMPLEMENTED` None exist in the repository today.
Cloudflare Stream / R2, SMS provider, KYC provider	`NOT SELECTED` Recommended, not procured, no accounts created, no pricing confirmed.
Latency assumption (`eu-west-3` over `af-south-1`)	`UNVERIFIED` Based on general routing, not on measurement from the target cities. Verify in P0.
Effort estimates in section M	`ESTIMATES` Two to three engineers assumed. Not commitments.
Regulatory position on holding funds	`UNRESOLVED` Requires local legal counsel. Outside what I can determine.
Existing repository prototype	`WORKING` Builds, lints, and was verified against a running server — but it is a directory demo with an in-memory store, not a foundation for this plan beyond its UI layer.

Q Open decisions I need from you

Answers to 1–4 change the plan materially. The rest can be settled during P0 without blocking anything.

1. **Regulatory posture.** Who is advising on whether Afrinext may hold and disburse user funds in the launch country? This gates P6 and could reshape the wallet's legal model.
2. **Launch scope.** Do you accept the digital-and-courses-first vertical, or do physical goods and delivery ship at launch? This reorders roughly nine weeks of work.
3. **iPay status.** Is the contract signed, and can you obtain the documentation and sandbox credentials listed in section P? If not, P3's manual rail becomes the launch payment method and we should plan for that openly.
4. **Referral levels.** Confirm single-level. If you want tiers, that needs the legal review in item 1 to cover it explicitly.
5. **Launch country and currency.** Niger and XOF assumed. Confirm, or name the first two.
6. **Team composition.** How many engineers, and what is their Prisma versus SQL experience? This settles the ORM question in section A, and it is the only reason that decision is still open.
7. **Hosting preference.** Managed and simple (Vercel + Neon, higher unit cost) or containers (Hetzner or Fly, cheaper, more operations)?
8. **Native app.** Is the PWA sufficient at launch, or is a store-distributed app required? It does not change the API design, but it changes how much weight the REST layer carries in P2 and P3.
9. **Cash on delivery.** Required at physical-goods launch, or can it be its own phase?
10. **KYC.** Which identity documents are practical to verify in the launch country, and is there a provider you already work with?

On approval, P0 and P1 are what I would start on — the monorepo, the schema, and the ledger with its property tests — because they are the parts that everything else is built on top of and the parts that are most expensive to change later. Nothing above is locked; tell me where you disagree and I will revise the blueprint before any code is written.